

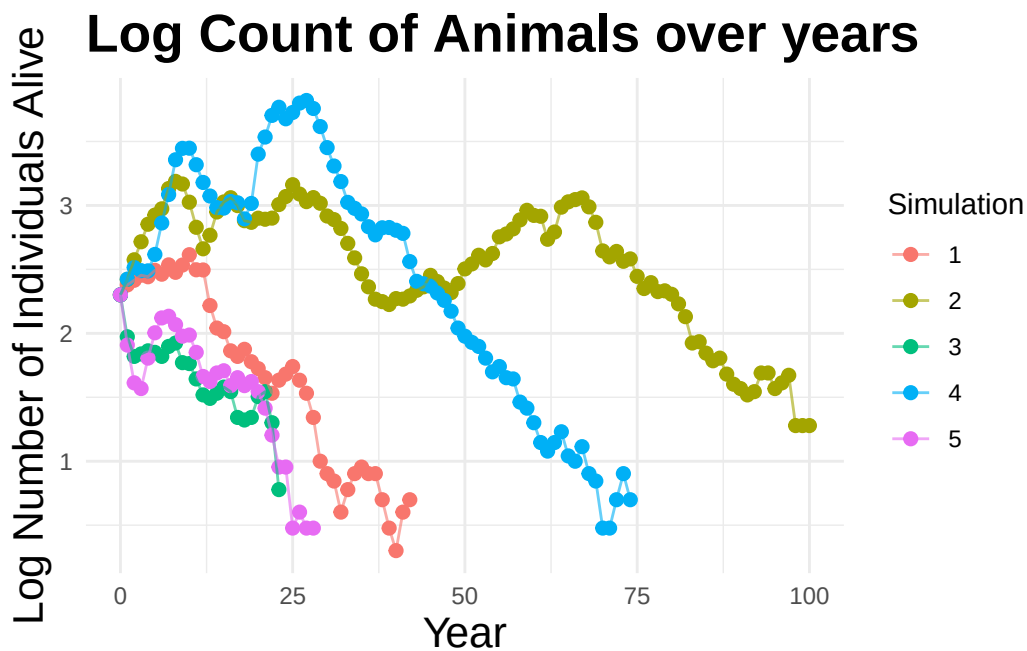
Time series exploration

```
pacman::p_load(dplyr, ggplot2)
set.seed(40)
```

```
source("../Main_function/calculate_trait&calculate_demo_v10.R")
```

```
final<-run_simulations( 5, #number of simulations
                        100, #years of the simulation
                        0.1, 0.5, #climate mean and s.d.
                        0, 1, 0.2, #trait int. mean and s.d.
                        0, 1, 0.2) #demo int. mean and s.d.
```

```
make_a_plot_sim(final)
```



We see that with these parameter all the 5 different simulations eventually go extinct.

The three events of reduction of the population are all happening in a fictional autumn/winter. The year of the animal is updated during spring after the reproduction event. Animal that are born in spring and died in autumn/winter of the same year are then counted.

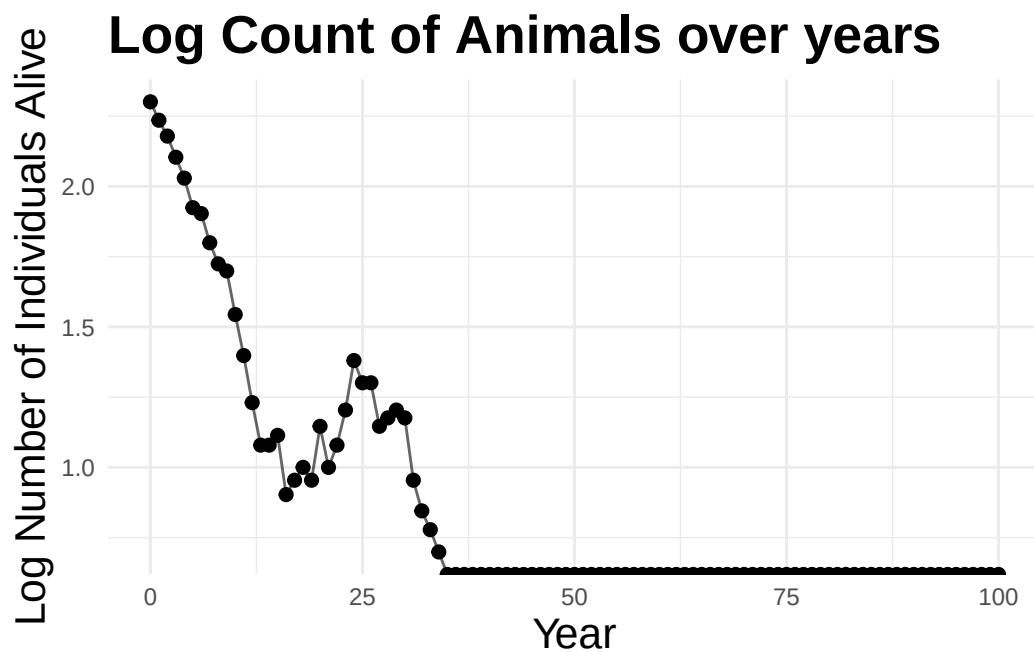
A warm year implies that newborns will have higher chances of survival and, once adults, will produce more offspring.

Conversely, in a cold year, the number of offspring depends solely on the fixed fertility values of the mothers alive in that year. Offspring born in cold years are less likely to survive in the following years and will produce fewer newborns themselves.

Konstant temperature analysis

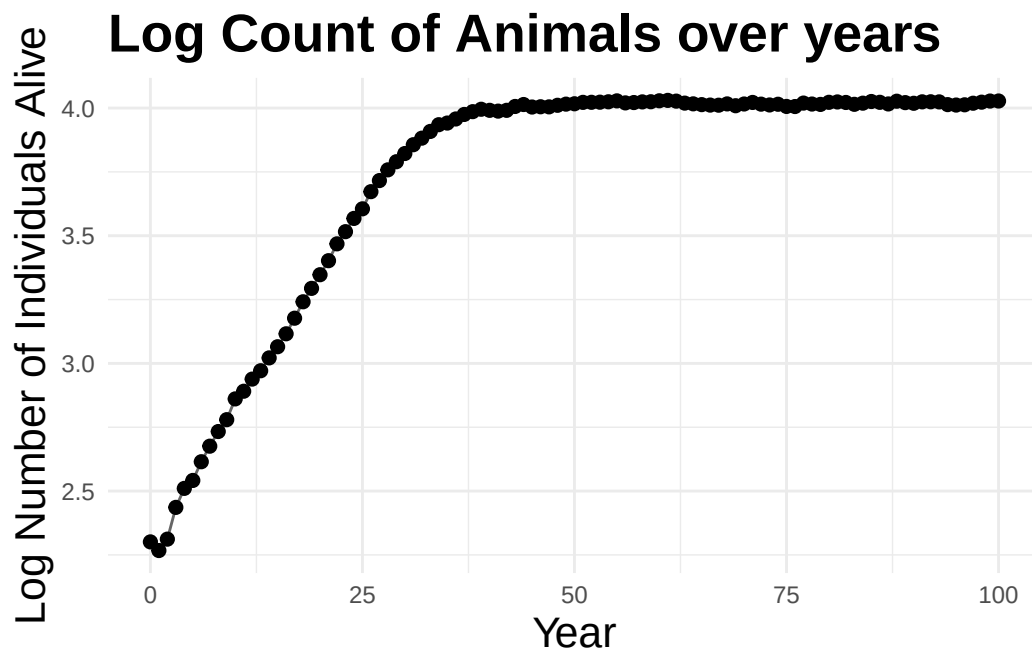
We set the temperature constant to a certain value and we leave the other parameter untouched.

```
k_01 <-readRDS("../Data&Plots/RDS/k_01.RDS")  
make_a_plot(k_01)
```



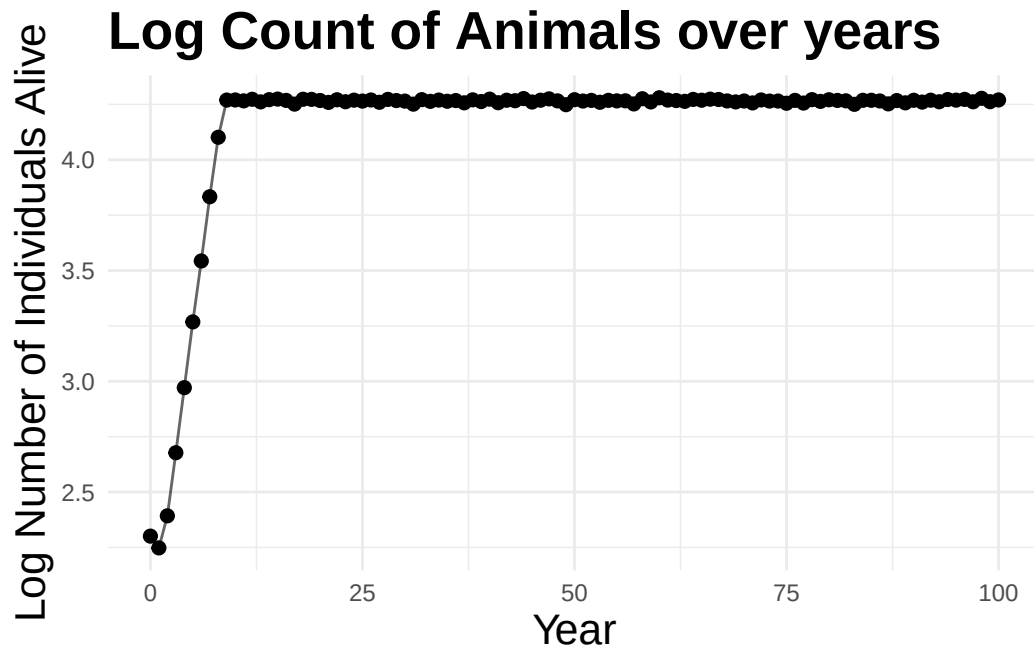
When temperature equals 0.1° the population goes extinct after 30 years

```
k_04 <-readRDS("../Data&Plots/RDS/k_04.RDS")  
make_a_plot(k_04)
```



On the other hand when we set the temperature is equal to 0.4° it reaches plateau to around 10^4 (10000) individuals. The favorable temperature is contrasted only by the density dependencies.

```
k_1 <-readRDS("../Data&Plots/RDS/k_1.RDS")  
make_a_plot(k_1)
```



Again with a constant temperature of 1° we reach plateau even quicker and its value is as expected a little higher than the previous case (around 30 000 individuals).